

Katrina Analysis

```
import pandas as pd
import numpy as np
from mlsynth import TSSC
import matplotlib.pyplot as plt

df = pd.read_csv("Total_Non_Farm_Employment_Katrina_2025.csv")

config = {
    "df": df,
    "outcome": "YoY_NFE",
    "treat": "Katrina",
    "unitid": "MSA Title",
    "time": "time",
    "display_graphs": False,
    "save": False,
    "counterfactual_color": ["red", "blue"]
}

Results = TSSC(config).fit()
y_obs = Results[3].raw_results['Vectors']['Observed Unit']
y_cf = Results[3].raw_results['Vectors']['Counterfactual']
post_periods = Results[3].raw_results['Fit']['Post-Periods']
pre_periods = Results[3].raw_results['Fit']['Pre-Periods']

split_date = "2005-08"
pre_index = pd.date_range(
    end=pd.Period(split_date, freq="M").start_time - pd.offsets.MonthEnd(1),
    periods=pre_periods,
    freq="ME"
)
post_index = pd.date_range(
    start=pd.Period(split_date, freq="M").start_time,
```

```

        periods=post_periods,
        freq="ME"
    )
    full_index = pre_index.append(post_index)

    fig, ax = plt.subplots(figsize=(10, 6))
    ax.plot(full_index, y_obs, label="New Orleans-Metairie, LA MSA (Observed)", linewidth=2.5, color="black")
    ax.plot(full_index, y_cf, label="Synthetic Control MSC(c) (Recommended)", linewidth=2.5, color="red")
    ax.axvline(x=pd.to_datetime(split_date), color="gray", linestyle=":", linewidth=2,
               label="Hurricane Katrina (Aug 2005)")
    ax.set_xlabel("Date", fontsize=11)
    ax.set_ylabel("Year-over-Year Employment Growth (%)", fontsize=11)
    ax.legend(fontsize=10)
    ax.grid(alpha=0.3)
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)
    plt.title("Causal Impact on YoY_NFE", fontsize=16, weight="bold")
    plt.tight_layout()
    plt.show()

```

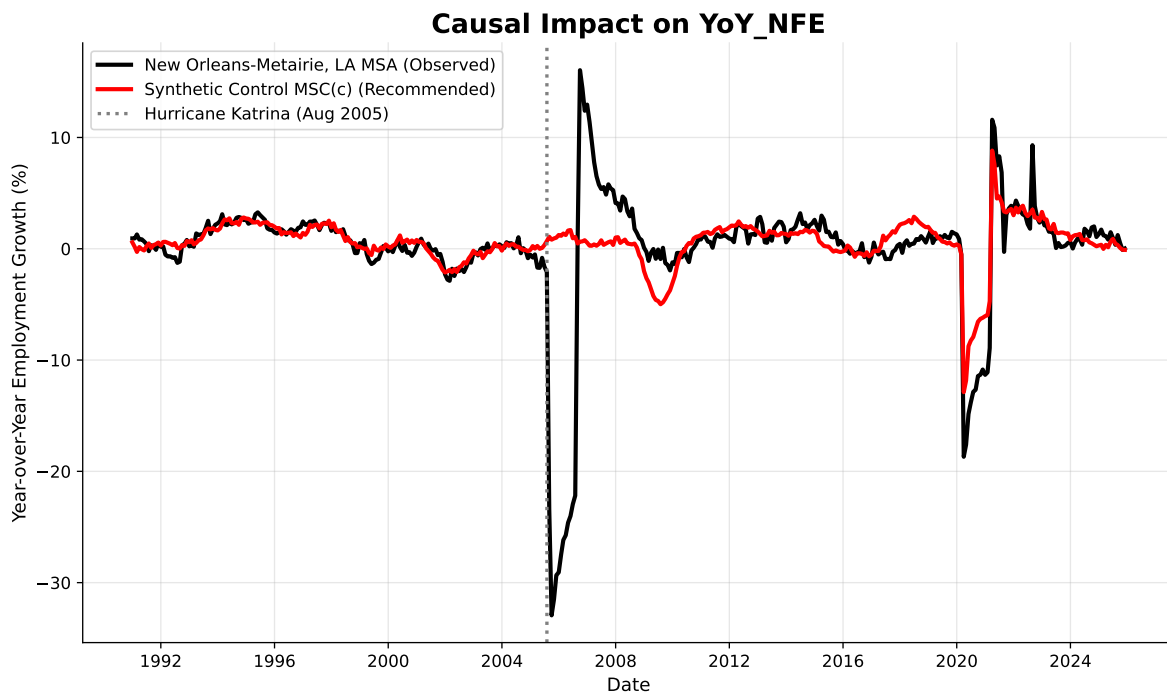


Figure 1: Observed vs Synthetic Control